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Studierichting: Informatica

Oefeningenreeks 7B nr. 1.3

$$w = \sin xyz, x = t, y = t^2, z = t^3$$

$$\frac{\partial w}{\partial s} = \begin{pmatrix} \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{pmatrix} \cdot \begin{pmatrix} \frac{\partial x}{\frac{ds}{\partial y}} \\ \frac{\frac{ds}{\partial y}}{\frac{ds}{\partial z}} \\ \frac{ds}{ds} \end{pmatrix}$$

$$\frac{\partial w}{\partial x} = yz \cos xyz, \frac{\partial x}{\partial s} = 1$$

$$\frac{\partial w}{\partial y} = xz \cos xyz, \frac{\partial y}{\partial s} = 2t$$

$$\frac{\partial w}{\partial z} = xy \cos xyz, \frac{\partial z}{\partial s} = 3t^2$$

$$\Rightarrow \frac{dw}{dt} = yz \cos xyz + 2txz \cos xyz + 3t^2 xy \cos xyz$$

$$= t^5 \cos t^6 + 2t^5 \cos t^6 + 3t^5 \cos t^6 = 6t^5 \cos t^6$$

References